Creality 3D Print Self Learning Skill

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Capstone II MIME 498

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Introduction

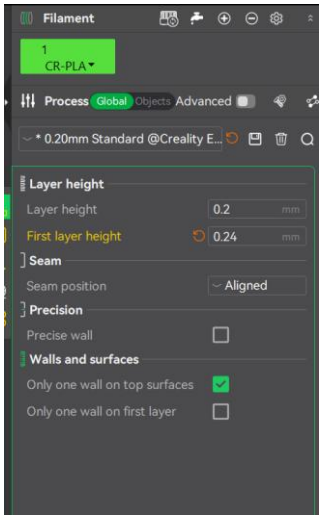
The Capstone II MIME 497 and 498 final project is aimed to exercise multiple disciplines used in the engineering industry. The Sumobot project is an example of this. The Sumobot is a robotics project, requiring rapid prototyping, advanced design, and creation of multiple product design iterations. Such requirements warrant the use of a 3D printer. 3D printers are exceptional for rapid prototyping, complex geometries, and are the most viable option for manufacturing by financial metrics. I chose to learn about 3D printing for my self-learning skill as I have never used this method of manufacturing before. Using a Creality Ender 3 V2 printer; I learned about 3D slicing software, support strategies and design considerations, PLA material properties and machine troubleshooting. I applied these techniques in the creation of the Sumobot's offensive mechanism.

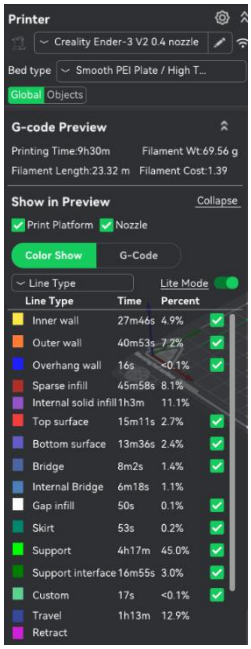
Creality Ender 3 V2

The Creality Ender 3 V2 printer is a 3-axis additive manufacturing machine. It takes a filament, usually made from PLA, and extrudes it through a nozzle. The nozzle has free movement along the x, y and z axis, and builds the part layer by layer from the build plate upward. The process starts by creating a CAD design file to model the part [3]. The design file is then exported as an STL file. Next, a program is used to create the build instructions for the part by converting the file into G-Code. Once the G-Code is uploaded to the printer, the part can be built.

3D Slicing Software

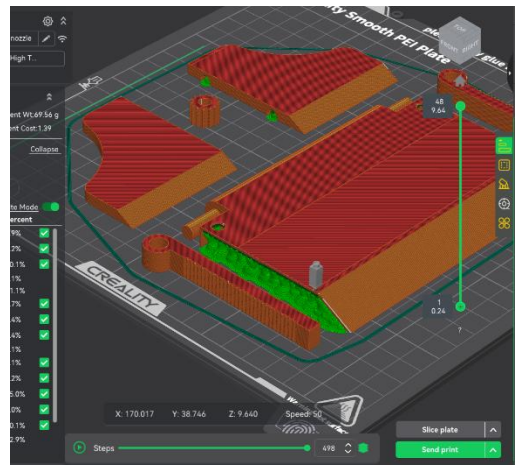
In order to use the 3D printer, a 3D slicing software is used so that the machine has build instructions. The 3D slicing software I have learned is the Creality Print slicing software. According to Creality Print, this software follows a lineage of slicing softwares. The original slicing software was created by Alessandro Ranellucci and the ReRap community which is called Slic3r [1]. The relevance here is that by learning this slicing software, I will be able to use this knowledge for similar software. To start slicing, you first position your part(s) on the virtual print bed. The part(s) must be oriented correctly in order to minimize print time, and support structures. Next, settings are selected which dictate the part creation. The Creality Print has abundant customization and settings which enable complete control over the creation of the part. The most important are the layer height, infill patterns, print speed, nozzle and bed temperature, wall thickness, and support structures.





Layer height corresponds to the vertical height of the part after one layer (one pass) of material. A part typically is not solid material; infill pattern is the geometry of the internal structure of the part. Print speed and temperatures are self-explanatory. Wall thickness is the thickness of the outer shell of the part. It has been important to understand how to use this software and the settings it provides.

Support Strategies and Design Considerations



The 3D printer builds the part from a bottom-up orientation. 3D printing hollow geometries or overhangs requires structural support to avoid part defects. The supports are removed from the finished part. It is important to design the part with respect to overhanging or hollow geometry to minimize the support needed [8]. This method of manufacturing requires different design considerations than casting, injection molding, machining, or other common plastic manufacturing processes.

Learning to orient the part, break the part into pieces, adding chamfers instead of overhangs, and use less material carries over to real world engineering practice. By using these skills I can create stronger, lighter parts and create them with minimal processing time for other projects.

PLA Material Properties

Polylactic Acid (PLA) is one of the most common materials used in 3D printing. PLA is used in rapid prototyping for its cost and melt temperature. Learning PLA material properties was important in

Mechanical properties		
	Typical value	Test method
MFR 210°C/2.16 kg	9.56 gr/10 min	ISO 1133
Tensile strength at yield	70 Mpa	ISO 527
Strain at yield	5 %	ISO 527
Strain at break	20 %	ISO 527
Tensile Modulus	3120 MPa	ISO 527
Impact strength-Charpy method 23°C	3.4 kJ/m²	ISO 179
Moisture absorption	1968 ppm	ISO 62

Thermal properties		
	Typical value	Test method
Melting temp.	115±35°C	ISO 11357
Vicat softening temp.	60 °C	ISO 306
Glass transition temp.	57 °C	ISO 11357

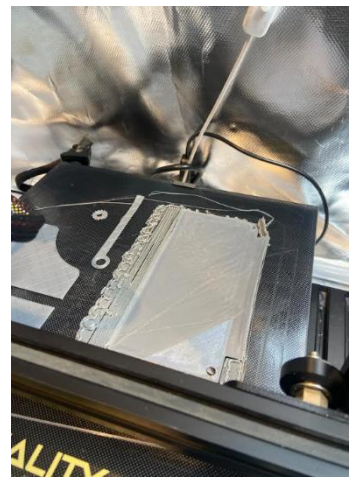
Printing settings	
Extruder temperature	190 °C - 220 °C
Bed temperature	65 °C
Speed	10-70 mm/s
Retraction speed	40 mm/s
Retraction distance	4 mm
Cooling fan	Yes
Minimum layer height	0.05 mm

the design process of the Sumobot. Compared to other common plastic materials such as polypropylene and polyethylene, PLA exhibits higher stiffness and tensile strength but significantly lower ductility and impact resistance. The Sumobot's frame is comprised solely of PLA material. The PLA mechanical properties are listed [7].

PLA is a strong but brittle material, and it is stiff rather than tough [9]. The main considerations for the robot were the low impact resistance, brittle failure modes, and heat sensitivity. To mitigate failure modes, we needed to apply rounded corners and fillets, reinforced sections, and ventilation to our design.

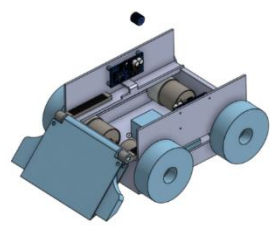
Troubleshooting

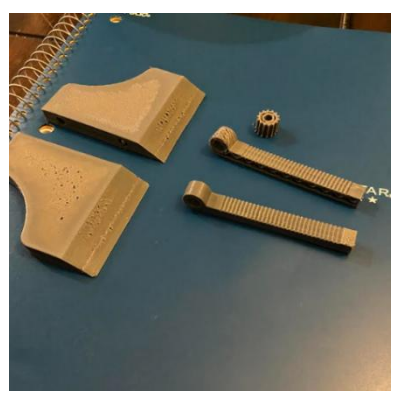
Throughout the project, troubleshooting played a critical role in improving print reliability and quality. Common issues such as poor bed adhesion, filament feeding inconsistencies, and print defects required adjustments to printer settings and setup. These challenges reinforced the importance of proper bed leveling, temperature control, and material handling. By gaining knowledge and experience through tutorials and online sources, I was able to resolve these issues and create more consistent and higher-quality prints. The experience and knowledge of working through issues whether machine malfunction [6], temperature settings, and leveling settings helped me create a better understanding of this manufacturing process



Project Application

The self-learned skill of 3D printing was applied to the design, iteration, and manufacturing of the Sumobot's offensive mechanism. CAD models were converted into physical parts for evaluation of fit, function, and structural performance, reducing development time compared to traditional methods. A key focus was part orientation and support strategy optimization within the slicing software. By orienting parts effectively, the need for supports was minimized, reducing material usage, print time, and post-processing. Design changes such as chamfers, fillets, and splitting components were used to improve manufacturability and reduce overhangs. Adjusting slicer parameters was essential to balance strength and efficiency. Structural components were printed with increased wall thickness and moderate-to-high infill to improve durability under impact, while less critical parts used lower infill to reduce weight. Knowledge of PLA material properties guided design decisions. Due to its brittle nature, components were reinforced and designed with rounded features to reduce stress concentrations. Print orientation was also selected to improve strength along load paths.





Conclusion

Learning 3D printing has been an extraneous process. I am still learning the optimization of 3D printed parts, and when certain design and manufacturing strategies are more feasible. Moving towards the future of my engineering career, I will take this knowledge along with me. 3D printers will remain the standard for rapid prototyping, and low-cost part building for years to come. I am thankful for this newly learned skill which I have learned outside of the engineering curriculum.

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